From an archived result to a useful contribution

Spencer Tropea | November-December 2026 | Conditional proposal

October establishes a replayable storage-state observation: 8,519,612 of 19,932,389 positive labels in the sampled stored chunks map to absent masked-CT objects. It does not establish physical registration, papyrus identity or ink.

Conditional November-December critical path - future work

Nov 2-6	Benchmark	Matched full-workflow comparison
Nov 9-13	Registration	Verified / failed / unresolved
Nov 16-20	User decision	One documented downstream action
Nov 23-30	Transfer	Frozen independent area; retain failures
Dec 1-11	Reproduction	Clean platforms; volunteer replay
Dec 14-18	Maintain / upstream / retire	Evidence-based release decision

Decision policy

Begin with approved, rights-reviewed reproducibility. Compare the existing tools fairly before claiming a missing function. Verify physical coordinates before interpreting material. Demonstrate a user decision before claiming practical impact.

> If baseline functionality is equivalent, contribute a narrow improvement upstream or stop duplication. If registration fails, publish the failure and retain only storage-state observations. Outcomes and external collaboration cannot be guaranteed.

Milestones with measurable gates

Window	Evidence to produce	Go / no-go decision
Oct 23-31	Approved source, tag and fixed-case replay; rights review	Keep private if approval or rights are unresolved
Nov 2-6	Matched full-workflow accuracy/agreement, time, memory, bytes and error comparison	Distinct value, upstream pivot or retire
Nov 9-13	Authoritative transforms, mask semantics and independent landmarks	Verified / failed / unresolved registration
Nov 16-20	One reproducible researcher decision and machine-readable export	Useful workflow or documented usability failure
Nov 23-30	Second area frozen before outcomes; complete denominator	Transfer finding or honest access failure
Dec 1-11	Clean Windows 3.13/Linux, schema, limits and volunteer replay request	Executed checks separated from untested promises
Dec 14-18	Technical note, changelog, feedback and maintenance decision	Maintain / merge upstream / retire
Dec 21-31	Documentation and follow-up buffer	No dependency on prize outcome or reviewer availability

R12 remains a separate exploratory branch. No ink analysis, letter interpretation or Gate 7F progression is on this critical path.

Opportunities for Collaboration

1. Review the nominal prediction-to-CT coordinate assumptions and point to authoritative transforms.

2. Reproduce H11 or one fixed cell from the original archived compressed bytes.

3. Test the audit on a different approved region or scan with its plan fixed before outcomes.

4. Identify a VC3D/villa or data-QA step where an inspection report would change a real decision.

5. Advise whether compact receipts, deterministic sampling or partition exports belong in phantom-audit upstream.

Commitment and constraints

I welcome scientific criticism and independent reproduction. The aim is to contribute something practical and maintainable, wherever it belongs in the existing toolchain. No role, endorsement, allocation or private access is requested.

Keep the CPU-first capture/replay path. Lock inputs, retain negative and unavailable results, and separate proposals from achieved milestones. AI-assisted software development remains disclosed and its outputs require tests and evidence review.

> A responsible year-end outcome may be a useful small patch, a limited storage-state tool, or a well-documented reason to stop. It is not a guarantee of discovery.

References

Official criteria: https://scrollprize.org/prizes Existing audit/cleanup: https://github.com/Schurkai/vesuvius-phantom-audit Data: https://scrollprize.org/data